Cross-Validation Robustness Test

Machine Learning Assignment 10

October 7, 2026

1 Introduction

In this assignment, we evaluate the robustness of the Decision Tree Classifier (previously built for the Loan Approval task) by applying k-fold cross-validation with $k = 10$. This technique helps us determine whether the model's performance is reliable across different subsets of data or if it was merely the result of a "lucky" train-test split.

2 Cross-Validation Results

The dataset was split into 10 folds, and the model was trained and evaluated 10 separate times. The accuracy for each fold is reported below:

Fold	Accuracy
1	0.9250
2	0.8500
3	1.0000
4	0.9750
5	0.9250
6	0.9750
7	0.9500
8	0.9250
9	0.9750
10	0.9000

Table 1: Accuracy per Fold for $k = 10$ Cross-Validation.

Summary Statistics

- **Mean Accuracy:** 0.9400 (94.00%)
- **Standard Deviation:** 0.0421 (4.21%)

3 Interpretation of Variance

What does a high standard deviation indicate?

In the context of cross-validation, the standard deviation measures how much the model's accuracy fluctuates depending on which specific data points are included in the training and testing sets.

A **high standard deviation** across folds would indicate that the model is **unstable or highly sensitive** to the training data. This typically happens for a few reasons:

1. **Overfitting:** The model might be memorizing specific patterns in certain folds that do not generalize well to other folds.
2. **Small Dataset Size:** There may not be enough data to consistently capture the true underlying distribution, causing wildly different performance metrics depending on the random split.

3. **Data Imbalance or Outliers:** Certain folds might accidentally contain a disproportionate amount of difficult-to-classify outliers or imbalanced classes.

In our specific experiment, the standard deviation is relatively low ($\sim$4.2%), indicating that the Decision Tree's performance is reasonably stable and reliable. If we had observed a standard deviation of 15-20%, we would conclude that reporting a single train-test accuracy score is misleading and that the model's predictive capability is untrustworthy in production.